AHP–TOPSIS Test Case

Module Name:-	AHP–TOPSIS Multi-Criteria Decision Visibility Engine
Test Case ID	WSV-001 to WSV-012
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Test Case Description	To check that the marketplace ranks listings by posting age, price drop, stock level and seller reliability.
Prerequisites:	1. API running with AHP-TOPSIS as the ranking (VISIBILITY_POLICY is not visibility-v4). 2. A feed of at least 5 other active listings from sellers with complete public profiles, with different age, price drop, stock and rating. 3. Scores are relative to the same feed. A lone listing scores 0.50.
Environmental Information:-	1. OS: Windows/Linux/Mac 2. Backend API on port 3000, buyer web on port 3001

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
Scenario:	Check the criteria weights (AHP).					
WSV-001	1. Run the AHP check script.	backend/src/ahp-test.ts	Weights are 0.351, 0.351, 0.189 and 0.109. They add up to 1. The consistency ratio is about 0.004, under the 0.10 limit.			
	2. Read the four weights and the consistency ratio.	Posting age, price drop, stock level, seller reliability				
Scenario:	Compare an old listing with a new one (Posting Age).					
WSV-002	1. Publish listing A with the same price, stock and seller rating as listing B.	A posted 5 hours ago, B posted just now	The older listing ranks above the newer one. Listing age is bigger-is-better, so produce that waited longer shows first.			
	2. Open the marketplace feed.	Marketplace feed				
	3. Compare the positions of A and B.	A and B				
Scenario:	Compare a lowered price with a full price (Price Drop).					
WSV-003	1. Take two listings with the same age, stock and seller rating.	Cabbage, 10 kg	The listing with the bigger peso drop (base price minus current price) ranks higher.			
	2. Lower the price of one with a price drop.	Base ₱100, now ₱80 vs ₱100				
	3. Open the marketplace feed.	Marketplace feed				
Scenario:	Compare a full-stock listing with a nearly empty one (Stock Level).					
WSV-004	1. Take two listings with the same age, price drop and seller rating.	Tomato	The listing with more stock ranks higher.			
	2. Set one to 20 kg and the other to 2 kg.	20 kg vs 2 kg				
	3. Open the marketplace feed.	Marketplace feed				

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
Scenario:	Compare a well-rated seller with a low-rated seller (Seller Reliability).					
WSV-005	1. Take two listings that are the same except for the seller.	Seller average order rating 4.8 vs 2.0	The listing from the higher-rated seller ranks higher. Only order service stars count, not product stars.			
	2. Open the marketplace feed.	Marketplace feed				
Scenario:	Publish a listing from a new seller with no reviews (Cold-Start).					
WSV-006	1. Log in as a seller with 0 reviews.	New seller account	The listing publishes with no error. The seller counts as 2.5 out of 5, and the rank comes from the other three criteria.			
	2. Publish a fresh listing.	Sitaw, 15 kg				
	3. Open the marketplace feed.	Marketplace feed				
Scenario:	View a feed with only one eligible listing (Single Listing).					
WSV-007	1. Hide or sell out every other listing.	Only 1 listing left	The one listing scores exactly 0.50 and still appears in the feed.			
	2. Open the marketplace feed.	Marketplace feed				
Scenario:	Hide listings from sellers who are not ready (Eligibility).					
WSV-008	1. Pick a seller whose profile setup is not finished, for example no map pin saved yet, with an active listing.	Seller with no map pin	The listings from sellers who are not ready do not appear. A seller with a complete, active profile still appears. After the seller is ready, the listing appears again.			
	2. Pick a seller whose email is not verified, and a seller who is paused or inactive, each with an active listing.	Unverified, paused and inactive sellers				
	3. Open the marketplace feed.	Marketplace feed				
	4. Finish the setup, verify the email or end the pause, then refresh the feed.	Same sellers				
Scenario:	Reserve the last kilograms of a listing (Sold Out).					
WSV-009	1. Reserve the remaining stock as a buyer.	Squash, last 3 kg	The listing leaves the feed. The product page says Sold out, ordering is disabled, and more reservations are refused.			
	2. Open the marketplace feed and the product page.	Marketplace feed, product page				
	3. Try to reserve again.	1 kg				
Scenario:	Publish two listings with the same criteria (Tie-Breaking).					
WSV-010	1. Publish two listings 5 minutes apart with the same stock, rating and history.	Carrots, then Okra	The order never changes. When scores tie, the newer listing is first, then the larger listing id.			
	2. Refresh the feed several times.	5 refreshes				

Test Case ID	Test Steps	Test Input	Expected Results	Actual Results	Status	Comments
Scenario:	Scroll the feed with more than 20 listings (Cursor Paging).					
WSV-011	1. Open the marketplace feed.	More than 20 active listings	Page 2 follows in score order with no repeats and no skipped listings. The old cursor gives 400 "Invalid marketplace cursor".			
	2. Load page 2 with the returned cursor.	nextCursor				
	3. Try a cursor older than 60 minutes.	Old nextCursor				
Scenario:	Switch the ranking back and forth (Safe Rollback).					
WSV-012	1. Set the policy in backend/.env, restart the API and refresh the feed.	VISIBILITY_POLICY=visibility-v4	With visibility-v4 the old formula runs. With no value, AHP-TOPSIS (visibility-ahptopsis-v1) returns. No errors either way.			
	2. Remove the setting, restart the API and refresh the feed.	No VISIBILITY_POLICY				